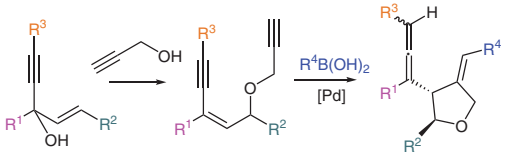
Synthesis Y.-M. Wang et al. Paper

Preparation of 3-Allenyl-4-methylenetetrahydrofurans

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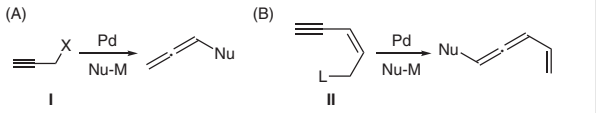
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Abstract A novel method for the preparation of 3-allenyl-4-methylenetetrahydrofurans via a Pd-catalyzed enyne cyclization reaction of propargyl 2-en-4-ynyl ethers under Suzuki coupling conditions is disclosed. The reaction proceeds via Pd activation of the terminal alkyne, cyclization, formation of the allenyl moiety, and reductive elimination, leading to the final product. This methodology offers several advantages, such as easy manipulation, mild reaction conditions, and easily accessible starting materials, providing a range of trisubstituted and 1,1-disubstituted allenyl(methylene)furan derivatives.

Key words allenes, furans, propargyl ethers, palladium, cyclization, Suzuki coupling

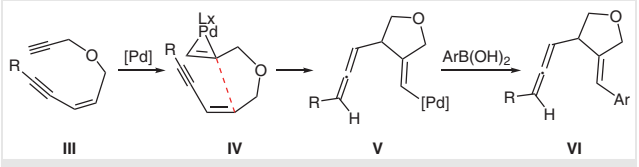
Allenyl heterocycles are useful synthetic intermediates due to the potential cyclization of heterocycles with the allenylic moiety to form polycyclic frameworks.[1,2] In addition, such allenyl heterocycles appear in natural products and pharmaceutical molecules.[2] Thus, development of synthetic methods for the construction of allenyl heterocycles and their applications have received much attention.[1–3]

Synthetic methodologies leading to allenyl compounds are well documented. Among them, Pd-catalyzed reaction of propargylic compounds **I** with nucleophilic organometallic or organoboron reagents to give the corresponding allene products is a known procedure (Scheme 1A).[4] In further applications, conjugated (*Z*)-2-en-4-ynes **II** behave similarly to provide the vinylallene products (Scheme 1B).[5]



Scheme 1 Conversion of propargylic compounds into allenes catalyzed by Pd

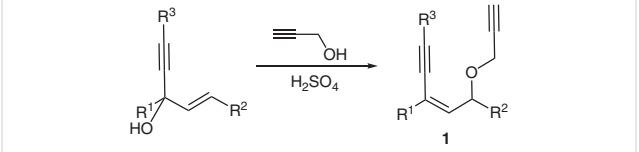
By adopting this concept, we imagined that a 2-en-4-yl propargyl ether **III** with a terminal alkynyl group moiety could be useful to prepare allenyl(methylene)furans **VI** (Scheme 2). Presumably, activation of the terminal alkyne by Pd gives **IV** as the primary intermediate.[6] Hopefully, an intramolecular addition to the enyne moiety followed by isomerization and protonation at the terminal allenyl carbon would take place to provide vinylpalladium species **V**. Aryl group transfer from the boron reagent to the Pd center and then reductive elimination would deliver the target framework **VI**.



Scheme 2 A proposed reaction pathway leading to allenyl(methylene)furans

1-En-4-yn-3-ols are useful synthetic intermediates and readily available from the addition of acetylides to chalcone molecules.[7] The substrates **1a–1r** were prepared by an S_N1-type ether formation between 1-en-4-yn-3-ols and propargyl alcohol under acidic conditions (Scheme 3). Various substituents at the R^1 to R^3 positions were prepared for this investigation.

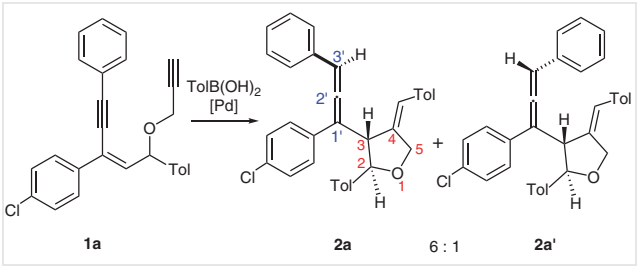
Investigation of the cyclization began with the reaction of **1a** with *p*-tolylboronic acid [$TolB(OH)_2$] in the presence of $Pd_2(dba)_3$ (5 mol%), $PCy_3{\cdot}HBF_4$ (40 mol%), and Na_2CO_3 in



Scheme 3 Preparation of propargyl ethers **1**

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Scheme 4 Reaction of **1a** with $TolB(OH)_2$ catalyzed by Pd

THF/MeOH solution under refluxing conditions (Scheme 4). To our delight, a pair of diastereomeric products, **2a** and **2a′**, were obtained. Compound **2a** was characterized by NMR, mass spectrometry, and X-ray crystallography. HRMS (ESI-TOF) gave a signal at $[M + H]^+$ m/z = 489.1950, consistent with the molecular formula of **2a**. The coupling constants $J_{(H2–H3)}$ for **2a** and **2a′** are 8.8 and 7.0 Hz, respectively, which are consistent with a *trans* arrangement of these protons.[8] In addition, the ^{13}C NMR spectrum of **2a** shows three signals at δ 207, 108, and 99 ppm, which are consistent with the signals of a trisubstituted allenyl moiety.[9] Nevertheless, X-ray crystallographic determination of **2a** does confirm the structural details including the relative stereochemistry (Figure 1).[10]

Optimization of the reaction conditions for a better production of **2a** was then studied (Table 1). The optimal conditions for this reaction are running the reaction of **1a** with *p*-tolylboronic acid (1.1 equiv) with the use of $Pd_2(dba)_3$ (5 mol%) and $PCy_3{\cdot}HBF_4$ (40 mol%) as the catalyst in the pres-

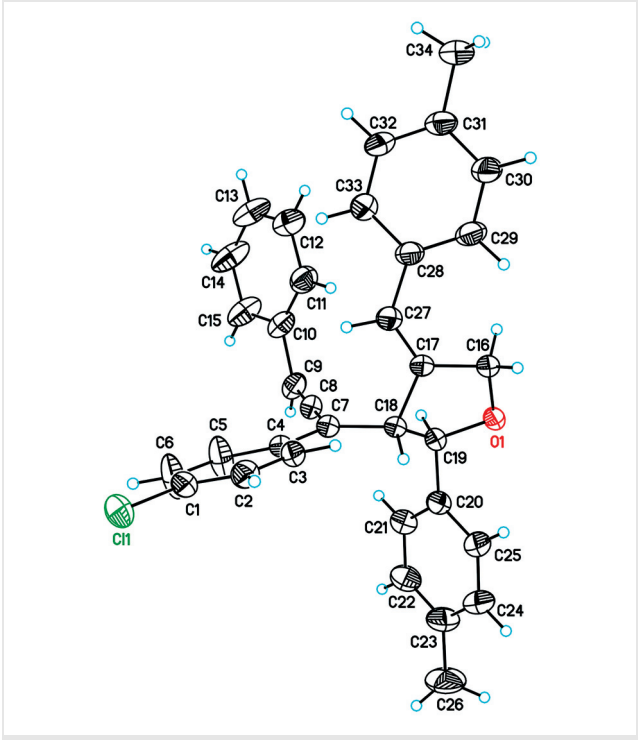


Figure 1 ORTEP plot of **2a** (30% probability ellipsoids)

Table 1 Reaction Optimization[a]

Entry	Base (equiv)	Solvent	Catalyst (mol%)	Yield (%)[b]
1	Na_2CO_3 (1.0)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	42
2	Na_2CO_3 (1.2)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	0
3	Na_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	81
4	Na_2CO_3 (0.6)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	0
5	K_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	0
6	Cs_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	0
7	NaOH (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	56
8	Na_2CO_3 (0.8)	MeCN/MeOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	46
9	Na_2CO_3 (0.8)	THF	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	0
10	Na_2CO_3 (0.8)	THF/^{t}BuOH (1:1)	$Pd_2(dba)_3$ (5), $PCy_3{\cdot}HBF_4$ (40)	0
11	Na_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $P^tBu_3{\cdot}HBF_4$ (40)	0
12	Na_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), dppe (20)	0
13	Na_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), DPEphos (20)	0
14	Na_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd_2(dba)_3$ (5), $Cy_2PCH_2CH_2PCy_2$ (20)	0
15	Na_2CO_3 (0.8)	THF/MeOH (1:1)	$Pd(PPh_3)_4$ (10)	0

[a] Reaction conditions: **1a** (0.1 mmol), base, catalyst, solvent (10 mL), stirred, 65 °C, 16 h, under N_2.
[b] NMR yields.

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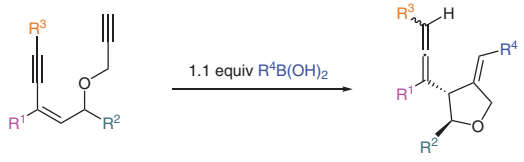
ence of Na_2CO_3 (0.8 equiv) in a mixed solvent of THF and MeOH at 65 °C for 16 hours (entry 3). Due to easy manipulation, we chose to use $PCy_3{\cdot}HBF_4$ as the ligand source instead of PCy_3. By screening various amounts of Na_2CO_3 (entries 1–4), we found that the use of 0.8 equivalent gave the best production of **2a** + **2a′**. Other bases, such as K_2CO_3 or Cs_2CO_3, did not provide the desired product at all (entries 5 and 6). Sodium hydroxide is another choice, but was not as good as Na_2CO_3 (entry 7). Carrying out the reaction in THF/MeOH was found to be the best medium and other solvent systems gave poor results (entries 8–10). Among various phosphine ligands, tricyclohexylphosphine appears to be the most suitable for the reaction while other phosphine ligands impeded the catalysis (entries 11–15).

With the optimized conditions, reactions of various substituted groups R^1, R^2, and R^3 in **1** with various aryl- or alkylboronic acids were examined (Table 2). Reactants **1** with various substituents R^1 reacted with phenylboronic acids smoothly under the catalytic conditions described above to give the allenylfuran products **2**, except for **1d** (R^1 = Me) (Table 2, entries 1–4). It appears that these substituents with different electronic natures affect the yield. When R^1 is a *p*-nitrophenyl group, the yield of **2b** dropped to 55% (entry 2). Unfortunately, compound **1d** with an alkyl substituent as R^1 did not provide the desired product **2d** at all (entry 4).

As for the boronic acids, phenylboronic acids with an electron-withdrawing or -donating substituent are suitable for the reaction, to give **2e** and **2f** in good yields (entries 5 and 6). However, reaction of **1a** with butylboronic acid gave an unexpected product, **2g** (R^4 = H), indicating that the butyl group underwent β-hydride elimination followed by reductive elimination to give the vinylidine moiety (entry 7). As for the R^2 group, reactants **1** with a *p*-fluorophenyl or 3,5-dimethoxyphenyl substituent are suitable substrates, giving **2h** and **2i** in 61% and 87% yield, respectively (entries 8 and 9). The reactivity of three substrates with different R^3 substituents was investigated. Both tolyl and butyl groups impede the formation of allenylfuran products (entries 10 and 12), whereas the reactant with R^3 as a dichlorophenyl group provided the expected product **2k** (entry 11). It appears that the nature of the R^3 substituent affects the reaction significantly.

Finally, we investigated the activity of substrates with R^3 as a trimethylsilyl (TMS) group. Compound **1p** (R^1 = *p*-ClC_6H_4, R^2 = *p*-MeC_6H_4, R^3 = TMS) reacted with $TolB(OH)_2$ under the catalytic conditions described above to yield **3h** in 78% yield (Scheme 5). The structure of **3h** was characterized by spectroscopic methods. The shortage of a ^1H NMR signal for the TMS group clearly shows that desilylation took place during the cyclization, rendering the 1,1-disubstituted allene product **3h**. Confirmation of this moiety

Table 2 Reaction Scope[a]



1 2

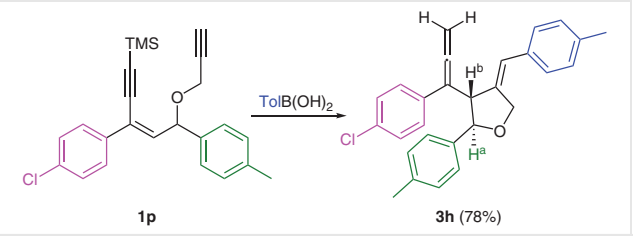
Entry	R^1	R^2	R^3	R^4	Product[b]
1	*p*-ClC_6H_4	*p*-MeC_6H_4	C_6H_5	*p*-MeC_6H_4	**2a** (78%, d.r. = 6:1)
2	*p*-$O_2NC_6H_4$	*p*-MeC_6H_4	C_6H_5	*p*-MeC_6H_4	**2b** (55%, d.r. = 3:1)
3	*p*-MeC_6H_4	*p*-MeC_6H_4	C_6H_5	*p*-MeC_6H_4	**2c** (84%, d.r. = 8:1)
4	Me	*p*-MeC_6H_4	C_6H_5	*p*-MeC_6H_4	**2d** (0%)
5	*p*-ClC_6H_4	*p*-MeC_6H_4	C_6H_5	*p*-HOC_6H_4	**2e** (74%, d.r. = 8:1)
6	*p*-ClC_6H_4	*p*-MeC_6H_4	C_6H_5	*p*-$MeCOC_6H_4$	**2f** (86%, d.r. = 6:1)
7	*p*-ClC_6H_4	*p*-MeC_6H_4	C_6H_5	Bu	**2g** (17%, d.r. = 4:1); R^4 = H
8	*p*-ClC_6H_4	*p*-FC_6H_4	C_6H_5	*p*-MeC_6H_4	**2h** (61%, d.r. = 8:1)
9	*p*-ClC_6H_4	3,5-$(MeO)_2C_6H_3$	C_6H_5	*p*-MeC_6H_4	**2i** (87%, d.r. = 5:1)
10	*p*-ClC_6H_4	*p*-MeC_6H_4	*p*-MeC_6H_4	*p*-MeC_6H_4	**2j** (0%)
11	*p*-ClC_6H_4	*p*-MeC_6H_4	3,5-$Cl_2C_6H_3$	*p*-MeC_6H_4	**2k** (75%, d.r. = 2:1)
12	*p*-ClC_6H_4	*p*-MeC_6H_4	Bu	*p*-MeC_6H_4	**2l** (0%)

[a] Reaction conditions: **1** (0.6 mmol), $Pd_2(dba)_3$ (0.03 mmol), $PCy_3{\cdot}HBF_4$ (0.24 mmol), $R^4B(OH)_2$ (0.66 mmol), Na_2CO_3 (0.48 mmol), THF/MeOH (1:1, 60 mL), 65 °C, 16 h, under N_2.
[b] Isolated yield given in parentheses.

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came from the ^{13}C NMR shifts at δ 209, 104, and 84 ppm corresponding to the 1,1-disubstituted allenylic carbons and ^{1}H NMR shifts for the methylidene group.[11] In addition, the coupling constant of $J_{(Ha–Hb)}$ = 7.5 Hz, similar to that for **2**, suggests the *trans* relationship of the substituents on the furan ring. The success in obtaining **3h** led us to prepare a series of this type of compound.



Scheme 5 Preparation of 1,1-disubstituted allenyl compound **3h**

As shown in Table 3, substrates **1** with various substituted phenyl rings R^1 and R^2 underwent cyclization followed by formation of the allenyl group to render the desired products **3** in good yields, except for **1l** (R^1 = *p*-MeC_6H_4, R^2 = *p*-MeC_6H_4) (Table 3, entry 3). This observation is unlike the formation of compound **2c** (Table 2, entry 3). Similar to **2g**, reaction of **1p** with butylboronic acid provided **3l** in low yield (14%) (Table 3, entry 12). It should be noted that **1q** (Figure 2) with an internal alkynyl moiety did not undergo the expected cascade cyclization and decomposed during the reaction. In addition, this reaction did not proceed with compound **1r** (Figure 2), with an allyl ether instead of propargyl ether, indicating the necessity of the alkynyl functionality.

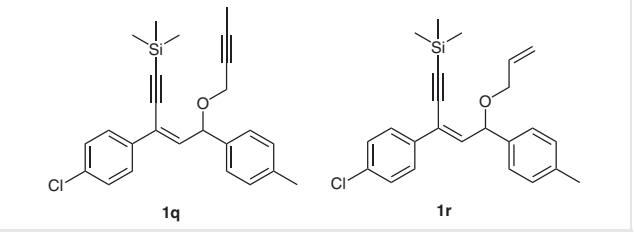
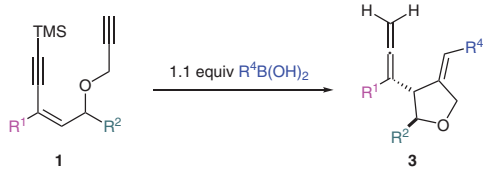


Figure 2 Structures of **1q** and **1r**

A possible reaction pathway leading to **2** is illustrated in Scheme 6. Activation of the terminal alkyne of **1** by Pd(0) results in the formation of **Int-1A**,[6a] which then undergoes nucleophilic attack at the enyne moiety to yield the furan ring, **Int-1B**, where R^2 is seated at the pseudo-equatorial position to avoid steric strain. This orientation of R^2 readily renders the stereochemistry of the furan ring in the final product. The allenyl anion in **Int-1B** is able to accept a proton from a solvent molecule to give **Int-1C**. Transmetalation of the R^4 group from $R^4B(OH)_2$ to Pd gives the intermediate **Int-1D**, which proceeds to undergo reductive elimination to yield the final product **2**. For **1** with R^3 as a TMS substituent, desilylation might take place during the reaction in the formation of **3**.

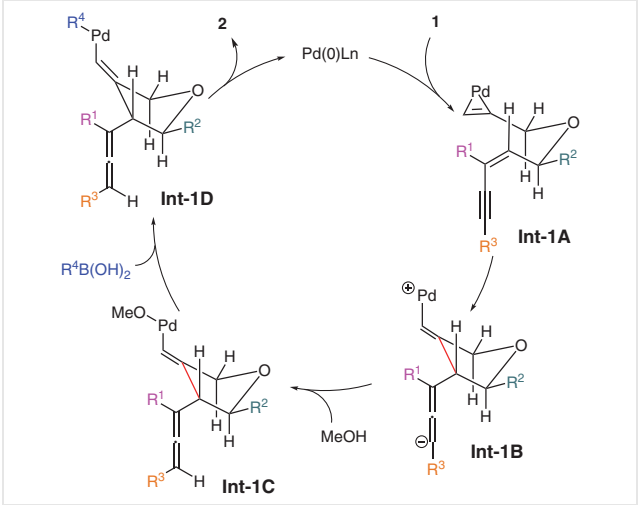
Table 3 Preparation of Compounds **3**[a]



Entry	R^1	R^2	R^4	Product[b]
1	C_6H_5	*p*-MeC_6H_4	*p*-MeC_6H_4	**3a** (64%)
2	*p*-FC_6H_4	*p*-MeC_6H_4	*p*-MeC_6H_4	**3b** (68%)
3	*p*-MeC_6H_4	*p*-MeC_6H_4	*p*-MeC_6H_4	**3c** (0%)
4	*p*-ClC_6H_4	*p*-FC_6H_4	*p*-MeC_6H_4	**3d** (80%)
5	*p*-ClC_6H_4	3,5-$(MeO)_2C_6H_3$	*p*-MeC_6H_4	**3e** (61%)
6	*p*-FC_6H_4	*p*-$MeOC_6H_4$	*p*-FC_6H_4	**3f** (70%)
7	*p*-ClC_6H_4	*p*-MeC_6H_4	C_6H_5	**3g** (83%)
8	*p*-ClC_6H_4	*p*-MeC_6H_4	*p*-MeC_6H_4	**3h** (78%)
9	*p*-ClC_6H_4	*p*-MeC_6H_4	*p*-$MeOC_6H_4$	**3i** (75%)
10	*p*-ClC_6H_4	*p*-MeC_6H_4	*p*-$MeCOC_6H_4$	**3j** (54%)
11	*p*-ClC_6H_4	*p*-MeC_6H_4	*p*-$O_2NC_6H_4$	**3k** (54%)
12	*p*-ClC_6H_4	*p*-MeC_6H_4	Bu	**3l** (14%); R^4 = H

[a] Reaction conditions: **1** (0.6 mmol), $Pd_2(dba)_3$ (0.03 mmol), $PCy_3·HBF_4$ (0.24 mmol), $R^4B(OH)_2$ (0.66 mmol), Na_2CO_3 (0.48 mmol), THF/MeOH (1:1, 60 mL), 65 °C, 16 h, under N_2.
[b] Isolated yield given in parentheses.

In summary, we have disclosed an efficient method to prepare 3-allenyl-4-methylenetetrahydrofurans **2** and **3** via the Pd-catalyzed enyne cyclization reaction of propargyl 2-



Scheme 6 Mechanistic pathway leading to **2**

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en-4-ynyl ethers **1** under Suzuki coupling conditions. This methodology offers several advantages such as easy manipulation, mild reaction conditions, and easily accessible starting materials. Further study to extend the application of allenyl(methylene)furans in synthetic utility is ongoing in our laboratory.

^{1}H and ^{13}C NMR spectra were recorded in $CDCl_3$ at 298 K on a 400 MHz spectrometer and referenced to TMS. All chemicals were commercially purchased and used without further purification. Flash chromatography was performed using silica gel 230–400 mesh. Melting points were determined on a Fargo MP-1D instrument. Unless otherwise noted, all reactions were performed under nitrogen atmosphere. Compounds **1** were prepared according to our previously reported method.[7]

Propargyl Ethers 1; General Procedure

In a 100-mL round-bottom flask was placed propargyl alcohol (1.01 g, 18 mmol) and H_2SO_4 (0.48 mL, 9 mmol) in THF (10 mL). A solution of 1-en-4-yn-3-ol (3 mmol) in THF (15 mL) was then syringed into the flask with stirring at room temperature for 2 h. The reaction was quenched with sat. aq $NaHCO_3$ and the mixture was extracted with ether (2 × 30 mL). The organic extracts were dried with $MgSO_4$ and concentrated. The residue was chromatographed on a silica gel column (hexane/EtOAc, 19:1).

(*Z*)-1-Chloro-4-(1-phenyl-5-(prop-2-yn-1-yloxy)-5-(*p*-tolyl)pent-3-en-1-yn-3-yl)benzene (1a)

Yield: 857.3 mg (72%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.61–7.50 (m, 4 H), 7.39–7.34 (m, 5 H), 7.29 (d, *J* = 8.7 Hz, 2 H), 7.16 (d, *J* = 7.8 Hz, 2 H), 6.48 (d, *J* = 8.9 Hz, 1 H), 5.84 (d, *J* = 8.9 Hz, 1 H), 4.16 (qd, *J* = 27.0, 2.4 Hz, 2 H), 2.39 (t, *J* = 2.4 Hz, 1 H), 2.30 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 137.8, 136.8, 136.4, 135.5, 134.1, 131.6, 129.3, 128.7, 128.5, 127.6, 126.7, 124.4, 122.8, 96.9, 85.5, 79.9, 78.9, 74.3, 55.6, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{22}ClO$: 397.1354; found: 397.1360.

(*Z*)-1-Methyl-4-(3-(4-nitrophenyl)-5-phenyl-1-(prop-2-yn-1-yloxy)pent-2-en-4-yn-1-yl)benzene (1b)

Yield: 709.1 mg (58%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.24 (d, *J* = 8.9 Hz, 2 H), 7.87 (d, *J* = 8.9 Hz, 2 H), 7.69–7.61 (m, 2 H), 7.48–7.44 (m, 5 H), 7.26 (d, *J* = 7.9 Hz, 2 H), 6.76 (d, *J* = 8.8 Hz, 1 H), 5.96 (d, *J* = 8.8 Hz, 1 H), 4.16 (qd, *J* = 32.0, 2.4 Hz, 2 H), 2.50 (t, *J* = 2.4 Hz, 1 H), 2.40 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 147.4, 143.3, 139.7, 138.2, 136.4, 131.6, 129.5, 129.1, 128.5, 127.0, 126.8, 123.7, 123.4, 122.4, 97.8, 84.7, 79.8, 78.9, 76.7, 74.6, 55.7, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{27}H_{21}NNaO_3$: 430.1414; found: 430.1392.

(*Z*)-4,4′-(5-Phenyl-1-(prop-2-yn-1-yloxy)pent-2-en-4-yne-1,3-diyl)bis(methylbenzene) (1c)

Yield: 903.6 mg (80%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.61–7.52 (m, 4 H), 7.39–7.33 (m, 5 H), 7.17–7.12 (m, 4 H), 6.46 (d, *J* = 9.0 Hz, 1 H), 5.86 (d, *J* = 9.0 Hz, 1 H), 4.16 (qd, *J* = 24.0, 2.4 Hz, 2 H), 2.38 (t, *J* = 2.4 Hz, 1 H), 2.33 (s, 3 H), 2.32 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 138.3, 137.7, 137.5, 135.3, 134.3, 131.7, 129.4, 129.2, 128.7, 128.5, 126.8, 126.4, 125.7, 123.2, 96.5, 86.4, 80.3, 79.1, 74.4, 55.7, 21.3, 21.2.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{28}H_{24}NaO$: 399.1719; found: 399.1696.

(*Z*)-1-Methyl-4-(3-methyl-5-phenyl-1-(prop-2-yn-1-yloxy)pent-2-en-4-yn-1-yl)benzene (1d)

Yield: 189.3 mg (21%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.59–7.52 (m, 2 H), 7.42–7.36 (m, 5 H), 7.23–7.19 (d, *J* = 7.8 Hz, 2 H), 5.90 (d, *J* = 9.1 Hz, 1 H), 5.71 (d, *J* = 9.1 Hz, 1 H), 4.16 (dd, *J* = 16.0, 1.9 Hz, 1 H), 4.27 (dd, *J* = 16.0, 1.9 Hz, 1 H), 2.44 (t, *J* = 1.9 Hz, 1 H), 2.39 (s, 3 H), 2.03 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 137.5, 136.7, 131.5, 129.2, 128.3, 126.9, 126.5, 123.2, 121.5, 94.2, 87.8, 80.1, 78.7, 74.0, 55.3, 23.1, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{22}H_{21}O$: 301.1587; found: 301.1592.

(*Z*)-1-Chloro-4-(5-(4-fluorophenyl)-1-phenyl-5-(prop-2-yn-1-yloxy)pent-3-en-1-yn-3-yl)benzene (1e)

Yield: 938.0 mg (78%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.67–7.55 (m, 4 H), 7.51–7.45 (m, 2 H), 7.42–7.36 (m, 3 H), 7.33 (d, *J* = 8.6 Hz, 2 H), 7.07 (t, *J* = 8.7 Hz, 2 H), 6.47 (d, *J* = 8.9 Hz, 1 H), 5.91 (d, *J* = 8.9 Hz, 1 H), 4.23 (qd, *J* = 25.0, 2.4 Hz, 2 H), 2.44 (t, *J* = 2.4 Hz, 1 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 163.7, 161.2, 135.9, 135.8, 135.3, 134.3, 131.6, 128.9, 128.6, 128.5, 128.4, 127.6, 124.9, 122.6, 115.6, 115.4, 97.1, 85.3, 79.8, 78.4, 74.5, 55.7.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –113.9.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{26}H_{19}ClFO$: 401.1103; found: 401.1110.

(*Z*)-1-(3-(4-Chlorophenyl)-5-phenyl-1-(prop-2-yn-1-yloxy)pent-2-en-4-yn-1-yl)-3,5-dimethoxybenzene (1f)

Yield: 810.6 mg (61%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.62–7.55 (m, 4 H), 7.42–7.36 (m, 3 H), 7.30 (d, *J* = 8.6 Hz, 2 H), 6.65 (d, *J* = 2.2 Hz, 2 H), 6.45 (d, *J* = 8.8 Hz, 1 H), 6.38 (t, *J* = 2.2 Hz, 1 H), 5.82 (d, *J* = 8.8 Hz, 1 H), 4.19 (qd, *J* = 26.0, 2.3 Hz, 2 H), 3.75 (s, 6 H), 2.41 (t, *J* = 2.3 Hz, 1 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 161.1, 142.3, 136.0, 135.4, 134.2, 131.6, 128.8, 128.5, 128.4, 127.6, 124.6, 122.7, 104.6, 99.9, 97.0, 85.5, 79.9, 79.1, 74.5, 55.7, 55.2.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{28}H_{24}ClO_3$: 443.1409; found: 443.1415.

(*Z*)-4,4′-(3-(4-Chlorophenyl)-5-(prop-2-yn-1-yloxy)pent-3-en-1-yne-1,5-diyl)bis(methylbenzene) (1g)

Yield: 949.3 mg (77%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.61 (d, *J* = 8.7 Hz, 2 H), 7.47 (d, *J* = 8.0 Hz, 2 H), 7.39 (d, *J* = 8.0 Hz, 2 H), 7.31 (d, *J* = 8.7 Hz, 2 H), 7.20–7.16 (m, 4 H), 6.49 (d, *J* = 8.9 Hz, 1 H), 5.87 (d, *J* = 8.9 Hz, 1 H), 4.19 (qd, *J* = 27.0, 2.4 Hz, 2 H), 2.41 (t, *J* = 2.4 Hz, 1 H), 2.39 (s, 3 H), 2.34 (s, 3 H).

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^{13}C NMR (100 MHz, $CDCl_3$): δ = 139.0, 137.8, 137.0, 136.1, 135.6, 134.1, 131.5, 129.3, 129.2, 128.5, 127.6, 126.7, 124.5, 119.8, 97.2, 84.9, 80.0, 79.0, 74.3, 55.6, 21.5, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{28}H_{23}ClNaO$: 433.1330; found: 433.1297.

(*Z*)-1,3-Dichloro-5-(3-(4-chlorophenyl)-5-(prop-2-yn-1-yloxy)-5-(*p*-tolyl)pent-3-en-1-yn-1-yl)benzene (1h)

Yield: 1159.8 mg (83%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.56 (d, *J* = 8.7 Hz, 2 H), 7.45 (d, *J* = 1.9 Hz, 2 H), 7.40–7.35 (m, 3 H), 7.33 (d, *J* = 8.7 Hz, 2 H), 7.21 (d, *J* = 7.9 Hz, 2 H), 6.58 (d, *J* = 8.7 Hz, 1 H), 5.83 (d, *J* = 8.9 Hz, 1 H), 4.19 (qd, *J* = 29.0, 2.4 Hz, 2 H), 2.47 (t, *J* = 2.4 Hz, 1 H), 2.36 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 138.0, 137.9, 136.5, 135.0, 134.9, 134.4, 129.7, 129.4, 128.9, 128.6, 127.5, 126.7, 125.5, 123.7, 94.8, 87.8, 80.0, 78.7, 74.4, 55.5, 21.1.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{20}Cl_3O$: 465.0575; found: 465.0580.

(*Z*)-1-Chloro-4-(1-(prop-2-yn-1-yloxy)-1-(*p*-tolyl)non-2-en-4-yn-3-yl)benzene (1i)

Yield: 859.3 mg (76%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, *J* = 8.5 Hz, 2 H), 7.35 (d, *J* = 7.9 Hz, 2 H), 7.27 (d, *J* = 8.5 Hz, 2 H), 7.17 (d, *J* = 7.9 Hz, 2 H), 6.38 (d, *J* = 8.9 Hz, 1 H), 5.74 (d, *J* = 8.9 Hz, 1 H), 4.16 (qd, *J* = 23, 2.3 Hz, 2 H), 2.51 (t, *J* = 7.0 Hz, 2 H), 2.43 (t, *J* = 2.3 Hz, 1 H), 2.34 (s, 3 H), 1.70–1.60 (m, 2 H), 1.58–1.47 (m, 2 H), 0.97 (t, *J* = 7.3 Hz, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 137.6, 137.1, 136.1, 135.1, 133.8, 129.2, 128.3, 127.5, 126.5, 124.9, 98.4, 80.1, 79.0, 73.9, 55.5, 30.7, 22.1, 21.1, 19.3, 13.5.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{25}H_{26}ClO$: 377.1667; found: 377.1660.

(*Z*)-Trimethyl(3-phenyl-5-(prop-2-yn-1-yloxy)-5-(*p*-tolyl)pent-3-en-1-yn-1-yl)silane (1j)

Yield: 1011.1 mg (94%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.68–7.64 (m, 2 H), 7.44–7.40 (m, 2 H), 7.38–7.31 (m, 3 H), 7.23 (d, *J* = 7.8 Hz, 2 H), 6.57 (d, *J* = 8.9 Hz, 1 H), 5.86 (d, *J* = 8.9 Hz, 1 H), 4.23 (qd, *J* = 23.0, 2.4 Hz, 2 H), 2.48 (t, *J* = 2.4 Hz, 1 H), 2.39 (s, 3 H), 0.37 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 137.6, 137.1, 137.0, 136.6, 129.2, 128.3, 128.2, 126.6, 126.2, 125.4, 102.3, 101.3, 80.1, 78.9, 74.1, 55.5, 21.1, –0.11.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{24}H_{26}NaOSi$: 381.1645; found: 381.1616.

(*Z*)-(3-(4-Fluorophenyl)-5-(prop-2-yn-1-yloxy)-5-(*p*-tolyl)pent-3-en-1-yn-1-yl)trimethylsilane (1k)

Yield: 1118.3 mg (99%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.70–7.60 (m, 2 H), 7.45 (d, *J* = 8.0 Hz, 2 H), 7.26 (d, *J* = 7.8 Hz, 2 H), 7.08 (t, *J* = 8.7 Hz, 2 H), 6.54 (d, *J* = 8.9 Hz, 1 H), 5.88 (d, *J* = 8.9 Hz, 1 H), 4.23 (qd, *J* = 26.0, 2.4 Hz, 2 H), 2.51 (t, *J* = 2.4 Hz, 1 H), 2.41 (s, 3 H), 0.40 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 163.9, 161.5, 137.7, 137.1, 136.9, 132.7, 129.3, 128.0, 127.9, 126.2, 124.2, 115.3, 115.1, 102.5, 101.2, 80.1, 79.0, 74.2, 55.5, 21.1, –0.11.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –113.5.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{24}H_{25}FNaOSi$: 399.1551; found: 399.1528.

(*Z*)-Trimethyl(5-(prop-2-yn-1-yloxy)-3,5-di(*p*-tolyl)pent-3-en-1-yn-1-yl)silane (1l)

Yield: 782.5 mg (70%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.49 (d, *J* = 8.1 Hz, 2 H), 7.36 (d, *J* = 8.0 Hz, 2 H), 7.16 (d, *J* = 8.1 Hz, 2 H), 7.12 (d, *J* = 8.0 Hz, 2 H), 6.46 (d, *J* = 9.0 Hz, 1 H), 5.79 (d, *J* = 9.0 Hz, 1 H), 4.17 (qd, *J* = 23, 2.4 Hz, 2 H), 2.42 (t, *J* = 2.4 Hz, 1 H), 2.34 (s, 3 H), 2.33 (s, 3 H), 0.29 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 138.1, 137.5, 137.2, 136.1, 133.8, 129.2, 128.9, 126.6, 126.1, 125.3, 101.9, 101.5, 80.1, 79.0, 74.0, 55.5, 21.1, 21.0, –0.11.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{25}H_{28}NaOSi$: 395.1802; found: 395.1780.

(*Z*)-(3-(4-Chlorophenyl)-5-(4-fluorophenyl)-5-(prop-2-yn-1-yloxy)pent-3-en-1-yn-1-yl)trimethylsilane (1m)

Yield: 1178.9 mg (99%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.57 (d, *J* = 8.2 Hz, 2 H), 7.50–7.43 (m, 2 H), 7.33 (d, *J* = 8.5 Hz, 2 H), 7.08 (t, *J* = 8.6 Hz, 2 H), 6.47 (d, *J* = 8.9 Hz, 1 H), 5.83 (d, *J* = 8.9 Hz, 1 H), 4.27 (d, *J* = 16 Hz, 1 H), 4.20 (d, *J* = 16 Hz, 1 H), 2.48 (t, *J* = 2.8 Hz, 1 H), 0.34 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 163.6, 161.2, 136.9, 135.7, 134.9, 134.2, 128.5, 128.3, 128.2, 127.5, 124.7, 115.5, 115.3, 103.1, 100.6, 79.7, 78.4, 74.4, 55.7, –0.20.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –114.0.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{23}H_{22}ClFNaOSi$: 419.1005; found: 419.1010.

(*Z*)-(3-(4-Chlorophenyl)-5-(3,5-dimethoxyphenyl)-5-(prop-2-yn-1-yloxy)pent-3-en-1-yn-1-yl)trimethylsilane (1n)

Yield: 1251.2 mg (95%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.56 (d, *J* = 8.4 Hz, 2 H), 7.32 (d, *J* = 8.4 Hz, 2 H), 6.66 (d, *J* = 2.2 Hz, 2 H), 6.46 (d, *J* = 8.9 Hz, 1 H), 6.42 (t, *J* = 2.2 Hz, 1 H), 5.78 (d, *J* = 8.9 Hz, 1 H), 4.23 (qd, *J* = 15, 2.0 Hz, 2 H), 3.82 (s, 6 H), 2.46 (t, *J* = 2.0 Hz, 1 H), 0.33 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 160.9. 142.2, 136.8, 135.0, 134.1, 128.4, 127.5, 124.4, 104.4, 102.7, 100.8, 99.9, 79.8, 79.0, 74.2, 55.7, 55.2, –0.17.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{25}H_{27}ClNaO_3Si$: 461.1310; found: 461.1283.

(*Z*)-(3-(4-Fluorophenyl)-5-(4-methoxyphenyl)-5-(prop-2-yn-1-yloxy)pent-3-en-1-yn-1-yl)trimethylsilane (1o)

Yield: 1130.5 mg (96%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.60–7.50 (m, 2 H), 7.36 (d, *J* = 8.7 Hz, 2 H), 6.98 (t, *J* = 8.7 Hz, 2 H), 6.87 (d, *J* = 8.7 Hz, 2 H), 6.42 (d, *J* = 8.9 Hz, 1 H), 5.72 (d, *J* = 8.9 Hz, 1 H), 4.16 (qd, *J* = 15, 2.3 Hz, 2 H), 3.78 (s, 3 H), 2.41 (t, *J* = 2.3 Hz, 1 H), 0.28 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 163.9, 159.4, 136.9, 132.7, 132.1, 127.9, 127.8, 127.1, 124.0, 115.3, 115.1, 113.9, 102.5, 101.1, 79.9, 78.7, 74.1, 55.4, 55.2, –0.15.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –113.7.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{24}H_{25}FNaO_2Si$: 415.1500; found: 415.1493.

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(*Z*)-(3-(4-Chlorophenyl)-5-(prop-2-yn-1-yloxy)-5-(*p*-tolyl)pent-3-en-1-yn-1-yl)trimethylsilane (1p)

Yield: 990.3 mg (84%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.64 (d, *J* = 8.5 Hz, 2 H), 7.49 (d, *J* = 7.9 Hz, 2 H), 7.39 (d, *J* = 8.5 Hz, 2 H), 7.29 (d, *J* = 7.8 Hz, 2 H), 6.62 (d, *J* = 8.9 Hz, 1 H), 5.94 (d, *J* = 8.9 Hz, 1 H), 4.31 (qd, *J* = 16, 2.1 Hz, 2 H), 2.54 (t, *J* = 2.1 Hz, 1 H), 2.44 (s, 3 H), 0.44 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 137.8, 137.6, 137.0, 135.2, 134.1, 129.4, 128.5, 127.6, 126.7, 124.2, 102.8, 101.0, 80.1, 79.0, 74.4, 55.6, 21.2, –0.04.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{24}H_{25}ClNaOSi$: 415.1255; found: 415.1235.

(*Z*)-(5-(But-2-yn-1-yloxy)-3-(4-chlorophenyl)-5-(*p*-tolyl)pent-3-en-1-yn-1-yl)trimethylsilane (1q)

Yield: 769.3 mg (63%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.54 (d, *J* = 8.7 Hz, 2 H), 7.38 (d, *J* = 8.0 Hz, 2 H), 7.29 (d, *J* = 8.7 Hz, 2 H), 7.19 (d, *J* = 8.0 Hz, 2 H), 6.53 (d, *J* = 8.8 Hz, 1 H), 5.75 (d, *J* = 8.9 Hz, 1 H), 4.14 (qq, *J* = 21, 2.3 Hz, 2 H), 2.35 (s, 3 H), 1.87 (d, *J* = 2.3 Hz, 3 H), 0.34 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 138.1, 137.6, 137.2, 135.2, 133.9, 129.2, 128.4, 127.5, 126.6, 123.6, 102.6, 101.1, 82.2, 78.9, 75.3, 56.3, 21.1, 3.69, –0.12.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{25}H_{27}ClNaOSi$: 429.1412; found: 429.1397.

(*Z*)-(5-(Allyloxy)-3-(4-chlorophenyl)-5-(*p*-tolyl)pent-3-en-1-yn-1-yl)trimethylsilane (1r)

Yield: 1019.1 mg (86%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.63 (d, *J* = 8.6 Hz, 2 H), 7.51 (d, *J* = 8.0 Hz, 2 H), 7.37 (d, *J* = 8.6 Hz, 2 H), 7.28 (d, *J* = 8.0 Hz, 2 H), 6.62 (d, *J* = 8.9 Hz, 1 H), 6.10 (ddt, *J* = 17.0, 10.0, 5.5 Hz, 1 H), 5.76 (d, *J* = 8.9 Hz, 1 H), 5.45 (dq, *J* = 17.0, 1.6 Hz, 1 H), 5.32 (dq, *J* = 10.0, 1.6 Hz, 1 H), 4.25–4.13 (m, 2 H), 2.43 (s, 3 H), 0.44 (s, 9 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 138.8, 137.9, 137.5, 135.3, 134.8, 134.0, 129.3, 128.5, 127.5, 126.5, 123.6, 117.0, 102.4, 101.4, 79.2, 69.4, 21.2, –0.01.

HRMS (ESI-TOF): *m*/*z* [M + Na]$^+$ calcd for $C_{24}H_{27}ClNaOSi$: 417.1412; found: 417.1372.

3-Allenyl-4-methylenetetrahydrofurans 2 and 3; General Procedure

A mixture of **1** (0.6 mmol), $Pd_2(dba)_3$ (27.5 mg, 0.03 mmol), $PCy_3{\cdot}HBF_4$ (88.4 mg, 0.24 mmol), Na_2CO_3 (50.8 mg, 0.48 mmol), and arylboronic acid (0.66 mmol) in THF/MeOH (1:1, 60 mL) in a flask equipped with reflux condenser was heated at 65 °C under nitrogen atmosphere for 16 h. Solvents were removed under reduced pressure and the residue was chromatographed on a silica gel column (hexane/EtOAc, 19:1).

3-(1-(4-Chlorophenyl)-3-phenylpropa-1,2-dien-1-yl)-4-((*Z*)-4-methylbenzylidene)-2-(*p*-tolyl)tetrahydrofuran (2a)

Yield: 228.9 mg (78%); d.r. = 6:1; light yellow oil.

Recrystallization gave **2a** as a pale yellow solid.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer, **2a**) = 7.37 (d, *J* = 8.6 Hz, 2 H), 7.33 (d, *J* = 8.0 Hz, 2 H), 7.28–7.21 (m, 7 H), 7.20–7.14 (m, 4 H), 7.03 (d, *J* = 8.0 Hz, 2 H), 6.67 (s, 1 H), 6.54–6.49 (m, 1 H), 5.11 (d, *J* = 14 Hz, 1 H), 4.93 (d, *J* = 8.8 Hz, 1 H), 4.85 (d, *J* = 14 Hz, 1 H), 3.96 (d, *J* = 8.8 Hz, 1 H), 2.37 (s, 3 H), 2.36 (s, 3 H); δ (minor isomer, **2a′**) = 7.37 (d, *J* = 8.6 Hz, 2 H), 7.33 (d, *J* = 8.0 Hz, 2 H), 7.28–7.21 (m, 7 H), 7.20–7.14 (m, 4 H), 7.03 (d, *J* = 8.0 Hz, 2 H), 6.65 (s, 1 H), 6.47–6.38 (m, 1 H), 5.03 (d, *J* = 15 Hz, 1 H), 4.93 (d, *J* = 8.8 Hz, 1 H), 4.85 (d, *J* = 15 Hz, 1 H), 4.07 (d, *J* = 7.0 Hz, 1 H), 2.37 (s, 3 H), 2.36 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer, **2a**) = 207.9, 142.7, 137.7, 137.1, 136.6, 134.0, 133.9, 133.3, 133.1, 129.2, 129.1, 128.7, 128.1, 127.9, 127.4, 127.1, 126.9, 126.4, 122.3, 108.4, 99.2, 83.9, 70.3, 55.1, 21.1, 21.0; δ (minor isomer, **2a′**) = 207.9, 142.7, 137.7, 137.1, 136.6, 134.0, 133.9, 133.3, 133.1, 129.2, 129.1, 128.7, 128.1, 127.9, 127.4, 127.1, 126.9, 126.4, 122.3, 108.4, 99.2, 83.9, 70.3, 55.1, 21.1, 21.0.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{34}H_{30}ClO$: 489.1980; found: 489.1950.

4-((*Z*)-4-Methylbenzylidene)-3-(1-(4-nitrophenyl)-3-phenylpropa-1,2-dien-1-yl)-2-(*p*-tolyl)tetrahydrofuran (2b)

Yield: 164.9 mg (55%); d.r. = 3:1; light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 8.15 (d, *J* = 8.9 Hz, 2 H), 7.58 (d, *J* = 8.0 Hz, 2 H), 7.35–7.24 (m, 7 H), 7.22–7.16 (m, 4 H), 7.03 (d, *J* = 8.1 Hz, 2 H), 6.79 (s, 1 H), 6.50 (q, *J* = 2.3 Hz, 1 H), 5.14 (d, *J* = 14 Hz, 1 H), 4.95–4.83 (m, 2 H), 4.03 (d, *J* = 8.8 Hz, 1 H), 2.39 (s, 3 H), 2.38 (s, 3 H); δ (minor isomer) = 8.17 (d, *J* = 7.8 Hz, 2 H), 7.60 (d, *J* = 7.8 Hz, 2 H), 7.35–7.24 (m, 7 H), 7.22–7.11 (m, 4 H), 7.07 (d, *J* = 8.1 Hz, 2 H), 6.77 (s, 1 H), 6.40 (q, *J* = 2.2 Hz, 1 H), 5.06 (d, *J* = 13 Hz, 1 H), 4.95–4.83 (m, 2 H), 4.14 (d, *J* = 7.8 Hz, 1 H), 2.38 (s, 3 H), 2.37 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 209.6, 146.7, 142.6, 142.3, 137.9, 136.9, 136.7, 133.7, 132.3, 129.2, 128.8, 127.9, 127.8, 127.3, 127.1, 126.4, 123.8, 122.6, 108.4, 99.9, 84.1, 70.3, 54.9, 21.1, 21.0; δ (minor isomer) = 208.9, 146.7, 142.7, 141.9, 138.0, 136.9, 136.6, 133.8, 132.2, 129.2, 128.8, 128.1, 127.3, 127.1, 126.8, 126.4, 123.9, 122.8, 108.4, 100.3, 84.7, 70.4, 54.4, 21.1, 21.0.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{34}H_{30}NO_3$: 500.2220; found: 500.2196.

4-((*Z*)-4-Methylbenzylidene)-3-((*R*)-3-phenyl-1-(*p*-tolyl)propa-1,2-dien-1-yl)-2-(*p*-tolyl)tetrahydrofuran (2c)

Yield: 236.2 mg (84%); d.r. = 8:1; light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.52–7.46 (m, 4 H), 7.41–7.35 (m, 4 H), 7.32–7.26 (m, 5 H), 7.24 (d, *J* = 8.0 Hz, 2 H), 7.17 (d, *J* = 8.1 Hz, 2 H), 6.78 (s, 1 H), 6.73–6.68 (m, 1 H), 5.24 (d, *J* = 14 Hz, 1 H), 5.12 (d, *J* = 8.8 Hz, 1 H), 5.00 (d, *J* = 14 Hz, 1 H), 4.14 (d, *J* = 8.8 Hz, 1 H), 2.48 (s, 3 H), 2.47 (s, 3 H), 2.45 (s, 3 H); δ (minor isomer) = 7.52–7.46 (m, 4 H), 7.41–7.35 (m, 4 H), 7.32–7.26 (m, 5 H), 7.24 (d, *J* = 8.0 Hz, 2 H), 7.17 (d, *J* = 8.1 Hz, 2 H), 6.75 (s, 1 H), 6.61–6.58 (m, 1 H), 5.17 (d, *J* = 12 Hz, 1 H), 5.12 (d, *J* = 7.7 Hz, 1 H), 5.00 (d, *J* = 12 Hz, 1 H), 4.14 (d, *J* = 7.7 Hz, 1 H), 2.52 (s, 3 H), 2.47 (s, 3 H), 2.44 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 207.8, 143.3, 137.6, 137.5, 137.2, 136.5, 134.3, 134.0, 132.6, 129.4, 129.3, 129.2, 128.7, 128.1, 127.0, 126.9, 126.6, 122.3, 109.3, 84.1, 70.5, 55.2, 21.3, 21.2, 21.1; δ (minor isomer) = 207.8, 143.3, 137.6, 137.5, 137.2, 136.5, 134.3, 134.0, 132.6, 129.4, 129.3, 129.2, 128.7, 128.1, 127.0, 126.9, 126.6, 122.3, 109.3, 84.1, 70.5, 55.2, 21.3, 21.2, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{35}H_{33}O$: 469.2526; found: 469.2502.

4-((*Z*)-(4-(1-(4-Chlorophenyl)-3-phenylpropa-1,2-dien-1-yl)-5-(*p*-tolyl)dihydrofuran-3(2*H*)-ylidene)methyl)phenol (2e)

Yield: 218.0 mg (74%); d.r. = 8:1; colorless oil.

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^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.46–7.34 (m, 5 H), 7.33–7.28 (m, 4 H), 7.25–7.17 (m, 4 H), 6.99 (d, *J* = 7.5 Hz, 2 H), 6.78 (d, *J* = 7.5 Hz, 2 H), 6.70 (s, 1 H), 6.50 (s, 1 H), 5.98 (s, 1 H), 5.11 (d, *J* = 14 Hz, 1 H), 4.98 (d, *J* = 8.5 Hz, 1 H), 4.88 (d, *J* = 14 Hz, 1 H), 4.00 (d, *J* = 8.5 Hz, 1 H), 2.40 (s, 3 H); δ (minor isomer) = 7.46–7.34 (m, 5 H), 7.33–7.28 (m, 4 H), 7.25–7.17 (m, 4 H), 6.99 (d, *J* = 7.5 Hz, 2 H), 6.78 (d, *J* = 7.5 Hz, 2 H), 6.70 (s, 1 H), 6.39 (s, 1 H), 5.80 (s, 1 H), 5.04 (d, *J* = 14 Hz, 1 H), 4.98 (d, *J* = 6.7 Hz, 1 H), 4.88 (d, *J* = 14 Hz, 1 H), 4.11 (d, *J* = 6.7 Hz, 1 H), 2.38 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 207.9, 154.8, 140.8, 137.9, 136.8, 133.9, 133.3, 133.1, 129.5, 129.3, 129.2, 128.7, 127.5, 126.9, 126.5, 122.2, 115.4, 108.4, 99.3, 84.1, 70.2, 54.9, 21.2; δ (minor isomer) = 207.9, 154.8, 140.8, 137.9, 136.8, 133.9, 133.3, 133.1, 129.5, 129.3, 129.2, 128.7, 127.5, 126.9, 126.5, 122.2, 115.4, 108.4, 99.3, 84.1, 70.2, 54.9, 21.2.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{33}H_{28}ClO_2$: 491.1773; found: 491.1780.

1-(4-((*Z*)-(4-(1-(4-Chlorophenyl)-3-phenylpropa-1,2-dien-1-yl)-5-(*p*-tolyl)dihydrofuran-3(2*H*)-ylidene)methyl)phenyl)ethan-1-one (2f)

Yield: 266.8 mg (86%); d.r. = 6:1; yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.97 (d, *J* = 8.4 Hz, 2 H), 7.43–7.36 (m, 4 H), 7.34–7.26 (m, 7 H), 7.25–7.19 (m, 4 H), 6.75 (s, 1 H), 6.66–6.61 (m, 1 H), 5.17 (d, *J* = 14 Hz, 1 H), 5.01 (d, *J* = 8.8 Hz, 1 H), 4.91 (d, *J* = 14 Hz, 1 H), 4.06 (d, *J* = 8.7 Hz, 1 H), 2.63 (s, 3 H), 2.41 (s, 3 H); δ (minor isomer) = 7.97 (d, *J* = 8.4 Hz, 2 H), 7.43–7.36 (m, 4 H), 7.34–7.26 (m, 7 H), 7.25–7.19 (m, 4 H), 6.72 (s, 1 H), 6.56–6.52 (m, 1 H), 5.10 (d, *J* = 16 Hz, 1 H), 5.01 (d, *J* = 7.9 Hz, 1 H), 4.91 (d, *J* = 14 Hz, 1 H), 4.12 (d, *J* = 7.9 Hz, 1 H), 2.63 (s, 3 H), 2.39 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 208.1, 197.3, 147.3, 141.4, 137.9, 136.8, 135.2, 133.8, 133.2, 133.1, 129.3, 128.9, 128.8, 128.7, 128.6, 128.2, 128.0, 127.8, 127.6, 127.4, 127.2, 126.9, 126.8, 126.4, 121.7, 108.4, 99.5, 84.3, 70.3, 55.4, 26.5, 21.2; δ (minor isomer) = 207.2, 197.3, 147.3, 141.4, 137.9, 136.8, 135.2, 133.8, 133.2, 133.1, 129.3, 128.9, 128.8, 128.7, 128.6, 128.2, 128.0, 127.8, 127.6, 127.4, 127.2, 126.9, 126.8, 126.4, 121.7, 108.4, 99.5, 84.3, 70.3, 54.7, 26.5, 21.2.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{35}H_{30}ClO_2$: 517.1929; found: 517.1935.

3-(1-(4-Chlorophenyl)-3-phenylpropa-1,2-dien-1-yl)-4-methylene-2-(*p*-tolyl)tetrahydrofuran (2g)

Yield: 40.7 mg (17%); d.r. = 4:1; light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.37–7.29 (m, 6 H), 7.28–7.20 (m, 5 H), 7.18 (d, *J* = 7.8 Hz, 2 H), 6.66 (s, 1 H), 5.13 (q, *J* = 2.4 Hz, 1 H), 5.09 (q, *J* = 2.4 Hz, 1 H), 4.93 (d, *J* = 8.8 Hz, 1 H), 4.78 (d, *J* = 13 Hz, 1 H), 4.59 (dq, *J* = 13.0, 2.4 Hz, 1 H), 3.78 (d, *J* = 8.8 Hz, 1 H), 2.37 (s, 3 H); δ (minor isomer) = 7.37–7.29 (m, 6 H), 7.28–7.20 (m, 5 H), 7.09 (d, *J* = 7.9 Hz, 2 H), 6.66 (s, 1 H), 5.07 (q, *J* = 2.4 Hz, 1 H), 5.03 (q, *J* = 2.4 Hz, 1 H), 4.90 (d, *J* = 8.4 Hz, 1 H), 4.74 (d, *J* = 14 Hz, 1 H), 4.52 (dq, *J* = 14.0, 2.4 Hz, 1 H), 3.87 (d, *J* = 8.4 Hz, 1 H), 2.35 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 207.3, 150.9, 137.7, 137.3, 134.0, 133.3, 133.0, 129.1, 128.9, 128.7, 128.6, 128.5, 127.9, 127.8, 127.5, 127.2, 126.9, 126.8, 126.4, 108.3, 105.9, 99.5, 85.0, 71.4, 53.1, 21.1; δ (minor isomer) = 206.6, 150.9, 137.7, 137.3, 134.0, 133.3, 133.0, 129.1, 128.9, 128.7, 128.6, 128.5, 127.9, 127.8, 127.5, 127.2, 126.9, 126.8, 126.4, 108.3, 105.9, 99.7, 85.0, 71.4, 52.9, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{24}ClO$: 399.1511; found: 399.1520.

3-(1-(4-Chlorophenyl)-3-phenylpropa-1,2-dien-1-yl)-2-(4-fluorophenyl)-4-((*Z*)-4-methylbenzylidene)tetrahydrofuran (2h)

Yield: 180.4 mg (61%); d.r. = 8:1; light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.44–7.35 (m, 4 H), 7.32–7.20 (m, 7 H), 7.18 (d, *J* = 7.9 Hz, 2 H), 7.09–7.01 (m, 4 H), 6.69 (s, 1 H), 6.53 (d, *J* = 2.4 Hz, 1 H), 5.12 (d, *J* = 14 Hz, 1 H), 4.93 (d, *J* = 9.0 Hz, 1 H), 4.59 (dt, *J* = 14.0, 2.4 Hz, 1 H), 3.91 (d, *J* = 9.0 Hz, 1 H), 2.36 (s, 3 H); δ (minor isomer) = 7.44–7.35 (m, 4 H), 7.32–7.20 (m, 7 H), 7.18 (d, *J* = 7.9 Hz, 2 H), 7.09–7.01 (m, 4 H), 6.67 (s, 1 H), 6.42 (d, *J* = 2.4 Hz, 1 H), 5.05 (d, *J* = 14 Hz, 1 H), 4.93 (d, *J* = 9.0 Hz, 1 H), 4.59 (dt, *J* = 14.0, 2.4 Hz, 1 H), 4.03 (d, *J* = 9.0 Hz, 1 H), 2.36 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 207.9, 163.8, 161.3, 142.3, 136.8, 135.9, 133.9, 133.8, 133.3, 133.2, 129.2, 128.8, 128.7, 128.2, 128.1, 128.0, 127.9, 127.8, 127.6, 126.9, 122.6, 115.5, 115.2, 108.1, 99.3, 83.4, 70.4, 55.5, 21.1; δ (minor isomer) = 207.1, 163.8, 161.3, 141.9, 136.8, 135.9, 133.9, 133.8, 133.3, 133.2, 129.2, 128.8, 128.7, 128.2, 128.1, 128.0, 127.9, 127.8, 127.6, 127.2, 122.8, 115.5, 115.2, 108.1, 99.7, 83.8, 70.4, 54.7, 21.1.

^{19}F NMR (376 MHz, $CDCl_3$): δ (major isomer) = –113.9; δ (minor isomer) = –114.0.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{33}H_{27}ClFO$: 493.1729; found: 493.1720.

3-(1-(4-Chlorophenyl)-3-phenylpropa-1,2-dien-1-yl)-2-(3,5-dimethoxyphenyl)-4-((*Z*)-4-methylbenzylidene)tetrahydrofuran (2i)

Yield: 279.3 mg (87%); d.r. = 5:1; light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.40 (d, *J* = 8.6 Hz, 2 H), 7.33–7.25 (m, 7 H), 7.17 (d, *J* = 8.0 Hz, 2 H), 7.00 (d, *J* = 8.0 Hz, 2 H), 6.70 (s, 1 H), 6.62 (d, *J* = 2.3 Hz, 2 H), 6.53 (q, *J* = 2.2 Hz, 1 H), 6.44 (t, *J* = 2.3 Hz, 1 H), 5.11 (d, *J* = 13 Hz, 1 H), 4.93 (d, *J* = 8.7 Hz, 1 H), 4.87 (dt, *J* = 14.0, 2.2 Hz, 1 H), 3.96 (d, *J* = 8.7 Hz, 1 H), 3.77 (s, 6 H), 2.36 (s, 3 H); δ (minor isomer) = 7.40 (d, *J* = 8.6 Hz, 2 H), 7.33–7.25 (m, 7 H), 7.17 (d, *J* = 8.0 Hz, 2 H), 7.05 (d, *J* = 8.0 Hz, 2 H), 6.65 (s, 1 H), 6.55 (d, *J* = 2.3 Hz, 2 H), 6.53 (q, *J* = 2.2 Hz, 1 H), 6.41 (t, *J* = 2.3 Hz, 1 H), 5.03 (d, *J* = 15 Hz, 1 H), 4.91 (d, *J* = 7.5 Hz, 1 H), 4.87 (dt, *J* = 15.0, 2.2 Hz, 1 H), 4.11 (d, *J* = 7.5 Hz, 1 H), 3.69 (s, 6 H), 2.36 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 208.1, 160.9, 142.8, 142.4, 136.7, 134.0, 133.9, 133.2, 133.1, 129.2, 128.7, 128.2, 127.9, 127.5, 126.9, 122.5, 108.6, 104.7, 104.2, 99.9, 99.2, 84.0, 70.4, 55.3, 55.2, 21.1; δ (minor isomer) = 207.2, 160.9, 142.5, 142.0, 136.7, 134.0, 133.9, 133.2, 133.1, 129.2, 128.7, 128.2, 127.9, 127.5, 127.0, 122.7, 108.6, 104.7, 104.2, 100.2, 99.6, 84.5, 70.3, 55.2, 54.5, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{35}H_{32}ClO_3$: 535.2034; found: 535.2002.

3-(1-(4-Chlorophenyl)-3-(3,5-dichlorophenyl)propa-1,2-dien-1-yl)-4-((*Z*)-4-methylbenzylidene)-2-(*p*-tolyl)tetrahydrofuran (2k)

Yield: 251.1 mg (75%); d.r. = 2:1; light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ (major isomer) = 7.41–7.22 (m, 7 H), 7.22–7.11 (m, 6 H), 7.09–7.06 (m, 2 H), 6.56 (s, 1 H), 6.53 (q, *J* = 2.3 Hz, 1 H), 5.12 (d, *J* = 14 Hz, 1 H), 4.93 (d, *J* = 8.4 Hz, 1 H), 4.87 (dt, *J* = 14.0, 2.3 Hz, 1 H), 4.00 (d, *J* = 8.4 Hz, 1 H), 2.37 (s, 6 H); δ (minor isomer) = 7.41–7.22 (m, 7 H), 7.22–7.11 (m, 6 H), 7.09–7.06 (m, 2 H), 6.52 (s, 1 H), 6.39 (q, *J* = 2.3 Hz, 1 H), 5.08 (d, *J* = 15 Hz, 1 H), 4.85 (d, *J* = 8.3 Hz, 1 H), 4.83 (dt, *J* = 15, 2.3 Hz, 1 H), 4.14 (d, *J* = 8.3 Hz, 1 H), 2.37 (s, 6 H).

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^{13}C NMR (100 MHz, $CDCl_3$): δ (major isomer) = 208.4, 142.4, 137.9, 136.9, 136.6, 135.3, 133.7, 133.1, 129.3, 129.2, 128.9, 128.2, 127.9, 127.4, 127.1, 126.3, 125.3, 125.1, 122.7, 110.1, 97.5, 84.0, 70.3, 55.1, 21.2, 21.1; δ (minor isomer) = 207.7, 142.0, 138.2, 136.9, 136.1, 135.3, 133.9, 133.2, 129.3, 129.2, 128.9, 128.2, 127.9, 127.4, 126.3, 125.3, 125.1, 109.8, 97.8, 84.3, 70.3, 54.1, 21.2, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{34}H_{28}Cl_3O$: 557.1201; found: 557.1210.

4-((*Z*)-4-Methylbenzylidene)-3-(1-phenylpropa-1,2-dien-1-yl)-2-(*p*-tolyl)tetrahydrofuran (3a)

Yield: 145.3 mg (64%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.43 (d, *J* = 7.3 Hz, 2 H), 7.36–7.31 (m, 4 H), 7.27–7.16 (m, 5 H), 7.09 (d, *J* = 8.1 Hz, 2 H), 6.40 (q, *J* = 2.3 Hz, 1 H), 5.23 (s, 2 H), 5.10 (d, *J* = 14 Hz, 1 H), 4.97–4.86 (m, 2 H), 3.91 (d, *J* = 7.6 Hz, 1 H), 2.39 (s, 3 H), 2.37 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 142.6, 137.5, 137.4, 136.5, 134.3, 129.1, 129.0, 128.4, 128.1, 126.9, 126.6, 122.2, 105.0, 83.9, 79.5, 70.3, 53.9, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{28}H_{27}O$: 379.2057; found: 379.2060.

3-(1-(4-Fluorophenyl)propa-1,2-dien-1-yl)-4-((*Z*)-4-methylbenzylidene)-2-(*p*-tolyl)tetrahydrofuran (3b)

Yield: 161.8 mg (68%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.39–7.32 (m, 2 H), 7.32 (d, *J* = 8.0 Hz, 2 H), 7.23–7.15 (m, 4 H), 7.08 (d, *J* = 8.0 Hz, 2 H), 7.02 (d, *J* = 8.8 Hz, 2 H), 6.37 (q, *J* = 2.3 Hz, 1 H), 5.22 (s, 2 H), 5.09 (d, *J* = 14 Hz, 1 H), 4.93–4.84 (m, 2 H), 3.86 (d, *J* = 7.9 Hz, 1 H), 2.39 (s, 3 H), 2.37 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.1, 163.0, 160.6, 142.4, 137.5, 137.4, 136.6, 134.1, 131.7, 129.2, 129.0, 128.2, 128.0, 126.3, 122.2, 115.4, 115.2, 104.1, 83.9, 79.7, 70.3, 54.2, 21.1.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –115.5.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{28}H_{26}FO$: 397.1963; found: 397.1970.

3-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-2-(4-fluorophenyl)-4-((*Z*)-4-methylbenzylidene)tetrahydrofuran (3d)

Yield: 200.1 mg (80%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.46–7.39 (m, 2 H), 7.37–7.29 (m, 4 H), 7.23 (d, *J* = 7.9 Hz, 2 H), 7.13–7.03 (m, 4 H), 6.39 (q, *J* = 2.3 Hz, 1 H), 5.27 (s, 2 H), 5.13 (d, *J* = 14 Hz, 1 H), 4.97–4.84 (m, 2 H), 3.82 (d, *J* = 8.5 Hz, 1 H), 2.40 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 163.7, 161.2, 141.8, 136.8, 136.2, 134.2, 133.9, 132.8, 129.2, 128.6, 128.1, 128.0, 127.9, 127.8, 122.5, 115.4, 115.2, 103.9, 83.5, 80.1, 70.4, 54.3, 21.2.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –114.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{23}ClFO$: 417.1416; found: 417.1425.

3-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-2-(3,5-dimethoxyphenyl)-4-((*Z*)-4-methylbenzylidene)tetrahydrofuran (3e)

Yield: 167.9 mg (61%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.29 (d, *J* = 8.6 Hz, 2 H), 7.24 (d, *J* = 8.6 Hz, 2 H), 7.15 (d, *J* = 7.9 Hz, 2 H), 7.02 (d, *J* = 7.9 Hz, 2 H), 6.54 (d, *J* = 2.2 Hz, 2 H), 6.37 (d, *J* = 2.2 Hz, 1 H), 6.31 (q, *J* = 2.3 Hz, 1 H), 5.24–5.14 (m, 2 H), 5.03 (d, *J* = 14 Hz, 1 H), 4.86–7.77 (m, 2 H), 3.79 (d, *J* = 6.4 Hz, 1 H), 3.74 (s, 6 H), 2.34 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 160.7, 142.9, 141.9, 136.7, 134.3, 134.0, 132.7, 129.2, 128.6, 128.0, 127.9, 122.4, 104.4, 104.0, 99.8, 83.9, 79.9, 70.4, 55.2, 54.1, 21.1.

HRMS (ESI): *m*/*z* [M + Na]$^+$ calcd for $C_{29}H_{27}ClNaO_3$: 481.1541; found: 481.1514.

4-((*Z*)-4-Fluorobenzylidene)-3-(1-(4-fluorophenyl)propa-1,2-dien-1-yl)-2-(4-methoxyphenyl)tetrahydrofuran (3f)

Yield: 174.1 mg (70%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.35–7.27 (m, 4 H), 7.13–7.07 (m, 2 H), 7.06–6.95 (m, 4 H), 6.87 (d, *J* = 8.7 Hz, 2 H), 6.31 (q, *J* = 2.3 Hz, 1 H), 5.23–5.14 (m, 2 H), 5.01 (d, *J* = 14 Hz, 1 H), 4.85–4.76 (m, 2 H), 3.83–3.75 (m, 4 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.0, 163.1, 162.7, 160.6, 160.3, 159.4, 143.2, 133.1, 132.2, 131.7, 129.7, 129.6, 128.2, 128.1, 127.6, 121.2, 115.5, 115.3, 113.8, 103.9, 83.9, 79.8, 70.1, 55.1, 54.0.

^{19}F NMR (376 MHz, $CDCl_3$): δ = –114.7, –115.3.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{23}F_2O_2$: 417.1661; found: 417.1668.

4-((*Z*)-Benzylidene)-3-(1-(4-chlorophenyl)propa-1,2-dien-1-yl)-2-(*p*-tolyl)tetrahydrofuran (3g)

Yield: 198.6 mg (83%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.47–7.26 (m, 9 H), 7.24–7.18 (m, 4 H), 6.46–6.41 (m, 1 H), 5.28 (s, 2 H), 5.15 (d, *J* = 14 Hz, 1 H), 4.99–4.89 (m, 2 H), 3.91 (d, *J* = 8.3 Hz, 1 H), 2.41 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 143.4, 137.6, 137.3, 136.9, 134.3, 132.7, 129.1, 128.6, 128.5, 127.8, 126.8, 126.3, 122.4, 104.2, 84.1, 80.1, 70.3, 53.9, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{24}ClO$: 399.1511; found: 399.1520.

3-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-4-((*Z*)-4-methylbenzylidene)-2-(*p*-tolyl)tetrahydrofuran (3h)

Yield: 193.2 mg (78%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.38–7.28 (m, 6 H), 7.21 (d, *J* = 8.6 Hz, 4 H), 7.10 (d, *J* = 8.1 Hz, 2 H), 6.39 (q, *J* = 2.3 Hz, 1 H), 5.30–5.20 (m, 2 H), 5.12 (d, *J* = 14 Hz, 1 H), 4.96–4.87 (m, 2 H), 3.88 (d, *J* = 7.5 Hz, 1 H), 2.41 (s, 3 H), 2.39 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 142.3, 137.6, 137.3, 136.6, 134.3, 134.1, 132.6, 129.2, 129.0, 128.5, 128.0, 127.8, 126.3, 122.2, 104.2, 83.9, 79.9, 70.3, 53.9, 21.1.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{28}H_{26}ClO$: 413.1667; found: 413.1670.

3-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-4-((*Z*)-4-methoxybenzylidene)-2-(*p*-tolyl)tetrahydrofuran (3i)

Yield: 193.0 mg (75%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.36–7.25 (m, 6 H), 7.17 (d, *J* = 8.0 Hz, 2 H), 7.11 (d, *J* = 8.0 Hz, 2 H), 6.93 (d, *J* = 8.0 Hz, 2 H), 6.32 (q, *J* = 2.4 Hz, 1 H), 5.23 (s, 2 H), 5.07 (d, *J* = 14 Hz, 1 H), 4.91–4.82 (m, 2 H), 3.88–3.79 (m, 4 H), 2.37 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 158.4, 140.9, 137.6, 137.3, 134.4, 132.6, 129.7, 129.4, 129.0, 128.5, 127.8, 126.3, 121.8, 113.9, 104.3, 83.9, 79.9, 70.3, 55.2, 53.9, 21.1.

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HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{28}H_{26}ClO_2$: 429.1616; found: 429.1620.

1-(4-((*Z*)-(4-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-5-(*p*-tolyl)-dihydrofuran-3(2*H*)-ylidene)methyl)phenyl)ethan-1-one (3j)

Yield: 156.1 mg (64%); yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.98 (d, *J* = 8.2 Hz, 2 H), 7.35–7.26 (m, 6 H), 7.24 (d, *J* = 8.2 Hz, 2 H), 7.17 (d, *J* = 7.9 Hz, 2 H), 6.41 (q, *J* = 2.1 Hz, 1 H), 5.26 (s, 2 H), 5.11 (d, *J* = 14 Hz, 1 H), 4.95–4.83 (m, 2 H), 3.88 (d, *J* = 8.4 Hz, 1 H), 2.62 (s, 3 H), 2.36 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.2, 197.3, 146.9, 141.5, 137.7, 136.9, 135.2, 134.1, 132.8, 129.1, 128.7, 128.6, 128.1, 127.7, 126.3, 121.5, 84.1, 80.3, 70.3, 54.1, 26.5, 21.1.

HRMS (ESI-TOF): *m*/*z* [M + H]$^+$ calcd for $C_{29}H_{26}ClO_2$: 441.1616; found: 441.1609.

3-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-4-((*Z*)-4-nitrobenzylidene)-2-(*p*-tolyl)tetrahydrofuran (3k)

Yield: 143.8 mg (54%); orange solid; mp 125–126 °C.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 8.18 (d, *J* = 8.6 Hz, 2 H), 7.27–7.20 (8 H), 7.12 (d, *J* = 7.9 Hz, 2 H), 6.37 (q, *J* = 2.3 Hz, 1 H), 5.26–5.16 (m, 2 H), 5.04 (d, *J* = 14 Hz, 1 H), 4.87–4.78 (m, 2 H), 3.83 (d, *J* = 8.4 Hz, 1 H), 2.31 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 209.1, 149.1, 146.1, 143.2, 136.5, 133.9, 132.9, 129.1, 128.7, 128.5, 127.6, 126.2, 124.3, 123.8, 120.6, 103.8, 84.2, 80.4, 70.2, 54.1, 21.1.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{27}H_{23}ClNO_3$: 444.1361; found: 444.1334.

3-(1-(4-Chlorophenyl)propa-1,2-dien-1-yl)-4-methylene-2-(*p*-tolyl)tetrahydrofuran (3l)

Yield: 27.1 mg (14%); light yellow oil.

^{1}H NMR (400 MHz, $CDCl_3$): δ = 7.31–7.22 (m, 6 H), 7.15 (d, *J* = 7.9 Hz, 2 H), 5.22 (s, 2 H), 5.03 (q, *J* = 2.3 Hz, 1 H), 4.94 (q, *J* = 2.3 Hz, 1 H), 4.86 (d, *J* = 8.7 Hz, 1 H), 4.76 (q, *J* = 13 Hz, 1 H), 4.57 (dq, *J* = 13.0, 2.3 Hz, 1 H), 3.61 (d, *J* = 8.7 Hz, 1 H), 2.34 (s, 3 H).

^{13}C NMR (100 MHz, $CDCl_3$): δ = 208.6, 150.4, 137.6, 137.3, 134.3, 132.5, 128.9, 128.4, 127.7, 126.2, 105.6, 103.7, 85.1, 80.2, 71.4, 52.1, 21.1.

HRMS (ESI): *m*/*z* [M + H]$^+$ calcd for $C_{21}H_{20}ClO$: 323.1198; found: 323.1190.

Crystallography

A crystal of **2a** suitable for X-ray determination was obtained by recrystallization. Data were collected at room temperature on a Bruker D8 Venture diffractometer. The structure was solved using the SHELXS-97 program[12] and refined using the SHELXL-97 program[13] by full-matrix least-squares on F^2 values.[10]

Conflict of Interest

The authors declare no conflict of interest.

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Supporting Information

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